Bochner's Theorem

Theorem 0.1 (Bochner)**.** *Given a locally compact Abelian group G, if $\phi : G \to \mathbb{C}$ is a continuous positive definite function, there exists a unique non-negative Borel measure $\hat{\mu}$ on $\hat{G}$ such that for all $x \in G$,*

$$\phi(x) = \int_{\hat{G}} \gamma(x) d\hat{\mu}(\gamma)$$

Proof. First of all, we claim that we can assume that $\phi(1) = 1$. To observe this, we note that for any $x_1, x_2, \ldots, x_n \in G$ and any $c_1, c_2, \ldots, c_n \in \mathbb{C}$, we have

$$\sum_{1 \leq i,j \leq n} c_i \overline{c_j} \phi(x_i x_j^{-1}) \geq 0.$$

Evaluating this for $n = 1$ and any c_1 and x_1 reveals that $\phi(1)$ must be real. Next, take $n = 2$, $x_1 = 1$, $x_2 = x$, $c_2 = 1$. Then, the above expression becomes $(1+|c_1|^2)\phi(1)+c_1\phi(x)+\overline{c_1}\phi(x^{-1}) \geq 0$. Evaluating this expression at $c_1 = 1$ and at $c_1 = i$ reveals that $\phi(x^{-1}) = \overline{\phi(x)}$. Finally, evaluate this expression by choosing c_1 such that $c_1\phi(x) = -|\phi(x)|$ to get that $\phi(1) \geq |\phi(x)|$ for all $x \in G$. This establishes that ϕ must be bounded, and thus, by multiplying a suitable constant, $\phi(1) = 1$.

Let $C_c(G)$ denote the space of continuous complex-valued functions on G with compact support. Pick any $f \in C_c(G)$ with support K and consider the function $g(x, y) = f(x)\overline{f(y)}\phi(xy^{-1})$ defined on $G \times G$ with support $K \times K$. As g is obtained by composition, addition, multiplication and conjugation of uniformly continuous functions, it is itself uniformly continuous on its support. (Here, we are using the fact that continuity along with compactness implies uniform continuity). This property will allow us to handle double integrals freely. Moving on, we consider:

$$\int_G \int_G g(x, y) d\mu(x) d\mu(y)$$

where μ is the normalised Haar measure of the group G. By the definition of Lebesgue integration, and from the above discussion, this integral is approximated by finite sums of the form:

$$\sum_{1 \leq i,j \leq n} g(x_i, x_j)\mu(E_i)\mu(E_j)$$

where $E_1, E_2, \ldots, E_n$ are disjoint measurable sets whose union is K and $x_i \in E_i$ for all i. Due to positive definiteness, the sum is always non-negative, and as a consequence, so is the above integral. Next, instead of taking $f \in C_c(G)$, if we take $f \in L^1(G)$, we can conclude that

$$\int_G \int_G f(x)\overline{f(y)}\phi(xy^{-1}) d\mu(x) d\mu(y) \geq 0$$

because any function in $L^1(G)$ is the limit of some sequence of functions from $C_c(G)$.

Next, we define a linear functional T_ϕ on $L_1(G)$ as

$$T_\phi(f) := \int_G \phi(x) f(x) d\mu(x)$$

, which is well-defined as ϕ is a bounded function. Particularly, as $|\phi(x)| \leq 1$ for all x, we get $|T_\phi(f)| \leq \|f\|$, and so, $\|T_\phi\| = 1$. Next, we use the notation $\tilde{h}$ to denote the function given by $\tilde{h}(x) = \overline{h(x^{-1})}$ for some function $h \in L_1(G)$, and define the commutator of any $f, h \in L_1(G)$ as $[f, h] = T_\phi(f * \tilde{h})$, where $*$ denotes the usual convolution operator. Thus, we have

$$[f, h] = \int_G \int_G f(x)\overline{h(y)}\phi(xy^{-1}) d\mu(x) d\mu(y).$$

From the above expression, it is clear that the commutator is linear in the first variable and anti-linear in the second, and that, $[f, h] = \overline{[h, f]}$. Also, by our initial discussion, we know that $[f, f] \geq 0$ for all f. These properties imply the Cauchy-Schwarz inequality for the commutator, which gives us

$$|[f, h]|^2 \leq [f, f][h, h].$$

Taking for h, the characteristic function of a neighbourhood V of 1 divided by $\mu(V)$, we have that

$$\begin{aligned}
[f, h] - T_\phi(f) &= \int_G \int_G f(x)\overline{h(y)}\phi(xy^{-1}) d\mu(x) d\mu(y) - \int_G \phi(x) f(x) d\mu(x) \\
&= \int_V \int_G \frac{f(x)}{\mu(V)} \phi(xy^{-1}) d\mu(x) d\mu(y) - \int_G \phi(x) f(x) d\mu(x) \\
&= \int_G \frac{f(x)}{\mu(V)} \int_V [\phi(xy^{-1}) - \phi(x)] d\mu(y) d\mu(x).
\end{aligned}$$

Next,

$$\begin{aligned}
[h, h] - 1 &= \int_G \int_G h(x)\overline{h(y)}\phi(xy^{-1}) d\mu(x) d\mu(y) - 1 \\
&= \frac{1}{\mu(V)^2} \int_V \int_V [\phi(xy^{-1}) - 1] d\mu(x) d\mu(y).
\end{aligned}$$

Choosing V to be a small enough compact set, the uniform continuity of ϕ implies along with the above equalities and the Cauchy-Schwarz inequality that:

$$|T_\phi(f)|^2 \leq [f, f] = T_\phi(f * \tilde{f}).$$

We'll use this inequality to get a functional on a different space altogether. Define $h = f * \tilde{f}$ and recursively, define $h^n = h^{n-1} * h$ for $n \geq 2$. An easy observation here is that $h = \tilde{h}$. Then, by a repeated application of the above inequality,

$$|T_\phi(f)|^2 \leq T_\phi(f * \tilde{f}) = T_\phi(h)$$

$$\leq T_\phi(h*h)^{1/2} = T_\phi(h^2)^{1/2}$$
$$\leq T_\phi(h^2*h^2) = \cdots \leq T_\phi(h^{2^n})^{1/2^n}$$
$$\leq \|h^{2^n}\|^{1/2^n}$$

By spectral radius formula, we know that $\|h^{2^n}\|^{1/2^n} \to \|\hat{h}\|_\infty$ as $n \to \infty$, where $\hat{h}$ denotes the Fourier transform of h. Thus,

$$|T_\phi(f)| \leq \|\hat{h}\|_\infty^{1/2} = \|\hat{f}\|_\infty$$

as Fourier transform of the convolution equals the product of the Fourier transforms of the two functions. The above inequality suggests that we can define a linear functional, which we will call T_ϕ by abuse of notation, on $A(\hat{G})$, the space of all Fourier transforms of functions in $L_1(G)$. Particularly, for any $\hat{f} \in A(\hat{G})$, we define $T_\phi(\hat{f}) := T_\phi(f)$. To show well-definedness, we see that if $\hat{f}_1 = \hat{f}_2$, then,

$$|T_\phi(f_1 - f_2)| \leq \|\hat{f}_1 - \hat{f}_2\|_\infty = 0,$$

and thus, $T_\phi(f_1) = T_\phi(f_2)$. Linearity is easy to check, and so, as $|T_\phi(\hat{f})| = |T_\phi(f)| \leq \|\hat{f}\|_\infty$, we get that T_ϕ is a valid bounded linear functional on $A(\hat{G})$ with respect to the supremum norm. As $A(\hat{G})$ is a subspace of the complex Banach space $C_0(\hat{G})$, the space of continuous functions on $\hat{G}$ vanishing at ∞, by Hahn-Banach Theorem, we get that T_ϕ can be extended to a functional on the entire space $C_0(\hat{G})$, such that the norm remains preserved. Now, as $\hat{G}$ is locally compact and Hausdorff too, by Riesz-Markov Representation Theorem, there exists a unique regular complex Borel measure $\hat{\mu}$ on $\hat{G}$ such that for all $h \in C_0(\hat{G})$,

$$T_\phi(h) = \int_{\hat{G}} h(\gamma^{-1}) d\hat{\mu}(\gamma),$$

such that the total variation $|\hat{\mu}|(\hat{G}) = \|T_\phi\|$ (which, we know, is equal to 1). Therefore, for all $f \in L_1(G)$,

$$\begin{aligned} T_\phi(f) &= T_\phi(\hat{f}) \\ &= \int_{\hat{G}} \hat{f}(\gamma^{-1}) d\hat{\mu}(\gamma) \\ &= \int_G \int_{\hat{G}} f(x)\gamma(x) d\hat{\mu}(\gamma) d\mu(x). \end{aligned}$$

We had defined $T_\phi(f) = \int_G \phi(x) f(x) d\mu(x)$, and so, for almost all x, we have

$$\phi(x) = \int_{\hat{G}} \gamma(x) d\hat{\mu}(\gamma).$$

As both sides of the above equation are continuous, we conclude that the equality holds for all $x \in G$. Next, for $x = 1$, the above equation gives that $1 = \phi(1) = \int_{\hat{G}} d\hat{\mu}(x) = \hat{\mu}(\hat{G}) \leq |\hat{\mu}|(\hat{G}) = 1$. Hence, equality holds everywhere, giving us in particular $\hat{\mu}(\hat{G}) = |\hat{\mu}|(\hat{G})$. Thus, μ can not take

negative values on any set.

Finally, we verify that the measure $\hat{\mu}$ is unique. Suppose we have two measures μ_1 and μ_2 on $\hat{G}$ that satisfy the above properties. Then, if $\mu' = \mu_1 - \mu_2$, we have

$$\int_{\hat{G}} \gamma(x) d\mu'(\gamma) = 0$$

for all $x \in G$. Reversing the arguments that we used above, we get that

$$\int_{\hat{G}} \hat{f}(\gamma) d\mu'(\gamma) = 0$$

for all $f \in L_1(G)$. Now, as the space $A(\hat{G})$ is dense in $C_0(\hat{G})$, again by the uniqueness clause of Riesz-Markov Representation Theorem, we conclude that $\mu' = 0$ or $\mu_1 = \mu_2$. Hence proved.

□